

Homework for Chap 8

With the effective mass approximation, do the following calculations:

1. Suppose the bandgap of GaAs at $T=300$ K is 1.42 eV and the electron effective mass is $0.062m_0$, calculate the density of states at energy of 1.52 eV and the effective density of states for nondegenerate electron gas in three dimensional (3D) GaAs.
2. For a GaAs quantum well with the well width of 3 nm, suppose the barrier height is infinite, calculate the first quantized energy E_1 for the 2-dimensional (2D) electrons and the corresponding electron density of states at this quantized energy E_1 .
3. Calculate the electron density of states for 3D electron gas at the energy E_1 , obtained in Q2, and compare the value with the density of states for 2D electrons at E_1 .

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1. According to the definition of density of electrons:

$$n = \int_{E_i}^{E_f} g(E) \cdot f(E) \cdot dE, \quad g(E) \text{ is the density of state}$$

With the formula with dimension $d=3$:

$$n = \int_{E_g}^{\infty} g_e \cdot \left(\frac{1}{2\pi^2}\right) \cdot \left(\frac{2m_e}{\hbar^2}\right)^{\frac{3}{2}} (E - E_g)^{\frac{1}{2}} \cdot \left(e^{\frac{E - E_F}{k_B T}} + 1\right)^{-1}$$

$$f(E) = \left(e^{\frac{E - E_F}{k_B T}} + 1\right)^{-1}$$

$$\text{so: } g(E) = g_e \cdot \frac{1}{2\pi^2} \cdot \left(\frac{2m_e}{\hbar^2}\right)^{\frac{3}{2}} (E - E_g)^{\frac{1}{2}}$$

$$g_e = g_s \cdot g_v, \text{ for GaAs, } g_s = 2, g_v = 1, \text{ so } g_e = 2$$

$$\text{For GaAs, } \frac{1}{m_{\text{eff}}} = \frac{1}{\hbar^2} \cdot \frac{\partial^2 E}{\partial k^2} \approx \frac{1}{0.062 m_0}$$

$$g(E) = 2 \cdot \frac{1}{2\pi^2} \cdot \left(\frac{2 \cdot 0.062 m_0}{\hbar^2}\right)^{\frac{3}{2}} (1.52 \times 1.6 \times 10^{-19} - 1.42 \times 1.6 \times 10^{-19})^{\frac{1}{2}}$$

$$g(E) = 2 \cdot \frac{1}{2\pi^2} \cdot \left(\frac{2 \cdot 0.062 m_0}{\hbar^2}\right)^{\frac{3}{2}} \cdot (1.52 \times 1.6 \times 10^{-19} - 1.42 \times 1.6 \times 10^{-19})^{\frac{1}{2}}$$

$$\approx 6.6 \times 10^{19} \cdot (\text{cm}^{-3} \cdot \text{eV}^{-1})$$

For effective density of states n_{eff} :

$$n = n_{\text{eff}} \cdot e^{-\left(\frac{E_g - E_F}{k_B T}\right)} = g_e \cdot \left(\frac{m_e \cdot k_B T}{2\pi \hbar^2}\right)^{\frac{3}{2}} \cdot e^{-\left(\frac{E_g - E_F}{k_B T}\right)}$$

$$\text{so: } n_{\text{eff}} = g_e \cdot \left(\frac{m_e \cdot k_B T}{2\pi \hbar^2}\right)^{\frac{3}{2}} \approx 3.9 \times 10^{17} (\text{cm}^{-3})$$

2. According to $E_{\text{tot}} = E_{\text{kinetic}} + E_{\text{quantized}}$:

$$E_n = \frac{\hbar^2 k^2}{2m_e} + \frac{\hbar^2 \pi^2 n^2}{2m_e \cdot l^2} \quad \text{for 2D}$$

$$E_{\text{quantized}} = \frac{\hbar^2 \pi^2 n^2}{2m_e \cdot l^2} \quad . \quad m_e = 0.062 m_0 \quad l = 3 \times 10^{-9} \text{ m}$$

For $n=1$:

$$E = \frac{\hbar^2 \pi^2}{2 \cdot m_e \cdot l^2} \approx 0.67 \text{ eV}$$

for $g_{2D}(E)$. As $g_{3D}(E) = g_e \cdot \left(\frac{1}{2\pi^2}\right) \cdot \left(\frac{2m_e}{\hbar^2}\right)^{\frac{d}{2}} \cdot (E - E_g)^{\frac{d}{2}-1}$, $d=3$,

When $d=2$:

$$g_{2D}(E) = g_e \cdot \left(\frac{1}{2\pi^2}\right) \cdot \frac{2m_e}{\hbar^2} \quad . \quad g_e = 2, \quad E = 0.67 \text{ eV}$$

$$g_{2D}(E) \approx 5.2 \times 10^{13} \text{ (cm}^{-2} \cdot \text{eV}^{-1})$$

3. With definition: ~~$g_{2D}(E)$~~ $g_{2D}(E) = \frac{1}{S} \cdot \frac{dN}{dE}$, $g_{3D} = \frac{1}{V} \frac{dN}{dE}$

$$\text{When } 2D \rightarrow 3D, \quad g_{3D} = \frac{1}{V} \frac{dN}{dE} = \frac{1}{S \cdot L} \frac{dN}{dE} = \frac{g_{2D}}{L}$$

Take $E = 0.67 \text{ eV}$ as confirmation:

$$g_{2D}(E) = 5.2 \times 10^{13} \text{ (cm}^{-2} \cdot \text{eV}^{-1}) \quad \frac{g_{2D}(E)}{L} \approx 1.72 \times 10^{20} \text{ (cm}^{-3} \cdot \text{eV}^{-1})$$

$$g_{3D}(E) \Rightarrow \frac{1}{2\pi^2} \left(\frac{2m_e}{\hbar^2}\right)^{\frac{3}{2}} \cdot \sqrt{0.67 \text{ eV}} \approx 1.708 \times 10^{20} \text{ (cm}^{-3} \cdot \text{eV}^{-1}) \quad . \text{ Correct}$$